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$$\int x^2 \sqrt{x-1} dx$$

$$\text{Stel } u = x - 1 \Rightarrow du = dx \text{ en } \Rightarrow \begin{cases} x = u + 1 \\ x^2 = (u + 1)^2 \end{cases}$$

$$= \int (u + 1)^2 \sqrt{u} du$$

$$= \int (u^2 + 2u + 1) u^{\frac{1}{2}} du$$

$$= \int u^{\frac{5}{2}} + 2u^{\frac{3}{2}} + u^{\frac{1}{2}} du$$

$$= \frac{2}{7}(x-1)^{\frac{7}{2}} + \frac{4}{5}(x-1)^{\frac{5}{2}} + \frac{2}{3}(x-1)^{\frac{3}{2}} + c$$

References